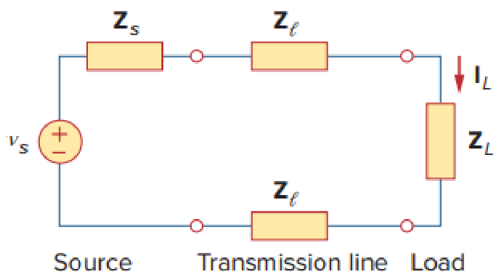
Chapter 9, Problem 93CP

Problem

A power transmission system is modeled as shown in Fig. 9.92. Given the source voltage and circuit elements

$\mathbf{V}_s = 115\underline{/0^\circ}$ V, source impedance
$\mathbf{Z}_s = (1 + j0.5)\ \Omega$, line impedance
$\mathbf{Z}_\ell = (0.4 + j0.3)\ \Omega$, and load impedance
$\mathbf{Z}_L = (23.2 + j18.9)\ \Omega$, find the load current $\mathbf{I}_L$.

Figure 9.92:



Step-by-step solution

Step 1 of 1

$$Z = Z_a + 2Z_b + Z_L$$

$$Z = (1+0.8+23.2) + j(0.5+0.6+18.9)$$

$$Z = 25 + j20$$

$$I_L = \frac{V_S}{Z} = \frac{115\angle 0^\circ}{32.02\angle 38.66^\circ}$$

$$I_L = 3.592\angle -38.66^\circ\ A \quad \textit{Answer}\ .$$